CIE 457 — Statistical Inference and Data Analysis

Final Project

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3.1 MLE and Model Formulation

We consider the linear model $y = X\beta + \varepsilon$. The objective is to find β that maximizes the likelihood of observing the data.

3.1.1 Likelihood Functions

Homoscedastic Case: $\varepsilon \sim N(0, \sigma^2 I)$

All observations have identical noise variance. The joint log-likelihood is:

$$\ell(\beta) = -n/2 \cdot \log(2\pi\sigma^2) - 1/(2\sigma^2) \cdot \|y - X\beta\|^2$$

Result: log-L at true $\beta = -150.300$ (Homo), -153.875 (Hetero).

Heteroscedastic Case: $\varepsilon \sim N(0, \Sigma)$, $\Sigma \neq \sigma^2 I$

Each observation has a different variance σ²_i. The log-likelihood now includes a weighted quadratic:

$$\ell(\beta) = -n/2 \cdot \log(2\pi) - 1/2 \cdot \log|\Sigma| - 1/2 \cdot (y-X\beta)^T\Sigma^{-1}(y-X\beta)$$

3.1.2 MLE Estimators

OLS from Homoscedastic MLE

Maximizing ℓ w.r.t. β is equivalent to minimizing $\|y-X\beta\|^2$. Setting $d\ell/d\beta = 0$:

$$\beta\hat{}\_OLS = (X^T X)^{-1} X^T y$$

WLS from Heteroscedastic MLE

Minimizing the weighted residuals $(X^T\Sigma^{-1}X)\beta = X^T\Sigma^{-1}y$ gives:

$$\beta\hat{}\_WLS = (X^T\Sigma^{-1}X)^{-1} X^T\Sigma^{-1} y$$

Estimator	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$	log-L Verification
True β	2.0000	1.5000	−0.8000	−150.300
OLS	2.0928	1.6907	−0.9721	−147.396 ≥ −150.300 ✓
WLS	2.1380	1.5742	−0.8287	MLE under hetero ✓

3.1.3 Model Assumptions and Distribution of $\hat{\beta}$

All Model Assumptions

- Linearity: $y = X\beta + \varepsilon$
- X is fixed and known (non-random design matrix)
- Noise is Gaussian: $\varepsilon \sim N(0, \sigma^2 I)$ or $\varepsilon \sim N(0, \Sigma)$
- $X^T X$ is invertible (no perfect multicollinearity)

Why $\hat{\beta}$ is a Random Variable

Substituting $y = X\beta + \varepsilon$ into the OLS formula:

$$\hat{\beta} = (X^T X)^{-1} X^T (X\beta + \varepsilon) = \beta + (X^T X)^{-1} X^T \varepsilon$$

X is fixed and β is constant ⇒ $\hat{\beta}$ is a linear function of the random ε. Every new sample produces a new ε and therefore a different $\hat{\beta}$. The five independent runs below confirm this:

Run	$\hat{\beta}_0$	$\hat{\beta}_1$	$\hat{\beta}_2$
1	1.9252	1.4593	−0.7344
2	2.0420	1.5623	−0.8165
3	1.9482	1.4262	−0.8433
4	2.0608	1.5262	−0.6703
5	2.1097	1.5490	−0.7826

Theoretical Distribution of $\hat{\beta}$

Since $\hat{\beta}$ is a linear transformation of a Gaussian, it is itself Gaussian:

Case	Distribution	Cov Trace
OLS (homo)	$N(\beta, \sigma^2(X^T X)^{-1})$	0.0341
WLS (hetero, correct)	$N(\beta, (X^T \Sigma^{-1} X)^{-1})$	0.0473
OLS (hetero, misspecified)	$N(\beta, (X^T X)^{-1} X^T \Sigma X (X^T X)^{-1})$	0.0592

3.2 Linear Regression under Homoscedastic Noise

Setup: n=100, β_true = [2.0, 1.5, −0.8], σ^2 = 1.0, N_sim = 5000 Monte Carlo runs.

3.2.1 Monte Carlo Results

Metric	β_0	β_1	β_2
True β	2.0000	1.5000	−0.8000
Empirical Mean	1.9988	1.5000	−0.8038
Theoretical Mean	2.0000	1.5000	−0.8000
Empirical Std	0.1026	0.1148	0.0999
Theoretical Std	0.1010	0.1174	0.1007

Empirical Mean ≈ True β ⇒ OLS is unbiased. Empirical Std ≈ Theoretical Std ⇒ variance formula $\sigma^2(X^TX)^{-1}$ is correct.

3.2.2 Empirical vs Theoretical Distribution

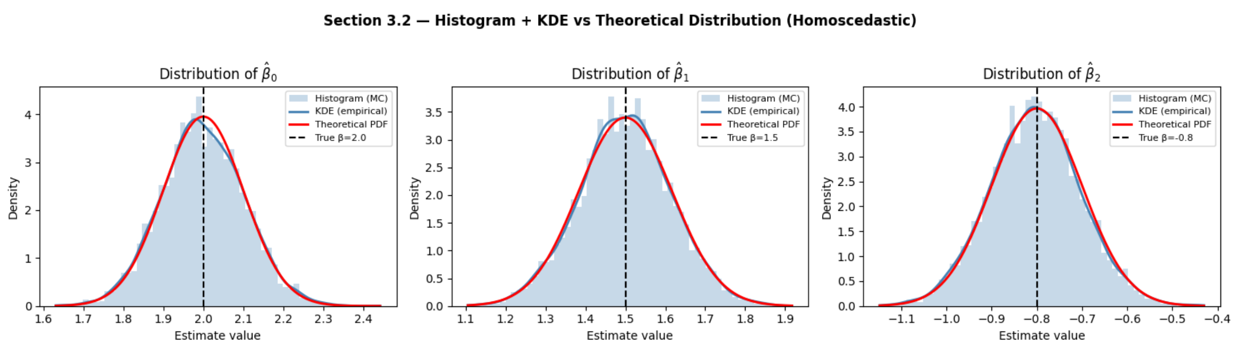

Figure 1 — Histogram + KDE (blue) vs Theoretical Gaussian PDF (red) for $\hat{\beta}_0$, $\hat{\beta}_1$, $\hat{\beta}_2$. Near-perfect overlap confirms $\hat{\beta} \sim N(\beta, \sigma^2(X^TX)^{-1})$. True β (black dashed) lies at the center of each distribution confirming unbiasedness.

3.3 Linear Regression under Heteroscedastic Noise

Noise structure: σ^2_i linearly spaced in [0.5, 3.0]. OLS (misspecified) and WLS (correct) are compared over N=5000 runs.

3.3.1 OLS vs WLS Results

Metric	OLS (misspecified)	WLS (correct)
Empirical Mean (β_0)	1.9978	1.9980
Empirical Std (β_0)	0.1340	0.1178
Empirical Std (β_1)	0.1540	0.1372
Empirical Std (β_2)	0.1344	0.1199
Cov Trace (theoretical)	0.0592	0.0473
Efficiency gain	—	1.25× lower variance

Both estimators are unbiased (Empirical Mean ≈ True β). However, WLS achieves 1.25× lower total variance by assigning weight $1/\sigma^2_i$ to each observation, exploiting the known noise structure.

3.3.2 OLS vs WLS Distribution

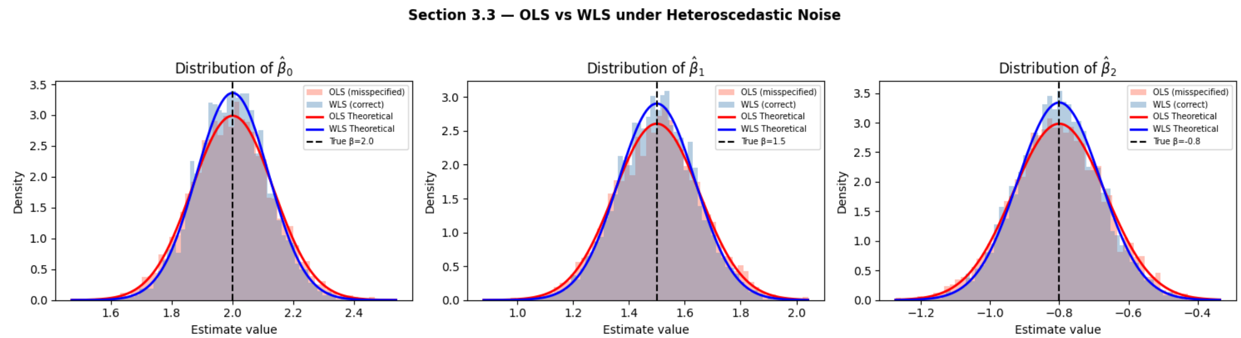

Figure 2 — OLS (red, wider) vs WLS (blue, narrower) under heteroscedastic noise. Both centered on true β (unbiased), but WLS is consistently narrower (more efficient).

3.4 Inference from Estimator Distribution

3.4.1 Confidence Intervals

Formula: β̂_j ± z_{0.025} · √(σ²[(XᵀX)⁻¹]_jj), where z = 1.96 = ppf(0.975).

	Estimate	Lower	Upper	True β	Inside?
β_0	2.0174	1.8195	2.2154	2.0000	✓
β_1	1.3935	1.1633	1.6237	1.5000	✓
β_2	−0.8554	−1.0528	−0.6581	−0.8000	✓

Coverage over 1000 samples: β_0 = 94.6%, β_1 = 94.6%, β_2 = 94.2% ≈ 95% ✓

3.4.2 Prediction Intervals

For x_new = [1.0, 0.5, −0.3]: y_hat = 2.971, y_true = 2.990.

Interval	Formula	Lower	Upper	Width
95% CI	y_hat ± 1.96√(σ² xᵀ(XᵀX)⁻¹x)	2.7189	3.2227	0.5037
95% PI	y_hat ± 1.96√(σ²(1+xᵀ(XᵀX)⁻¹x))	0.9947	4.9469	3.9522

PI is 7.85× wider than CI because it includes future noise σ^2 = 1.0 in addition to estimation uncertainty (0.0165). As n → ∞, CI → 0 but PI stays wide (irreducible noise).

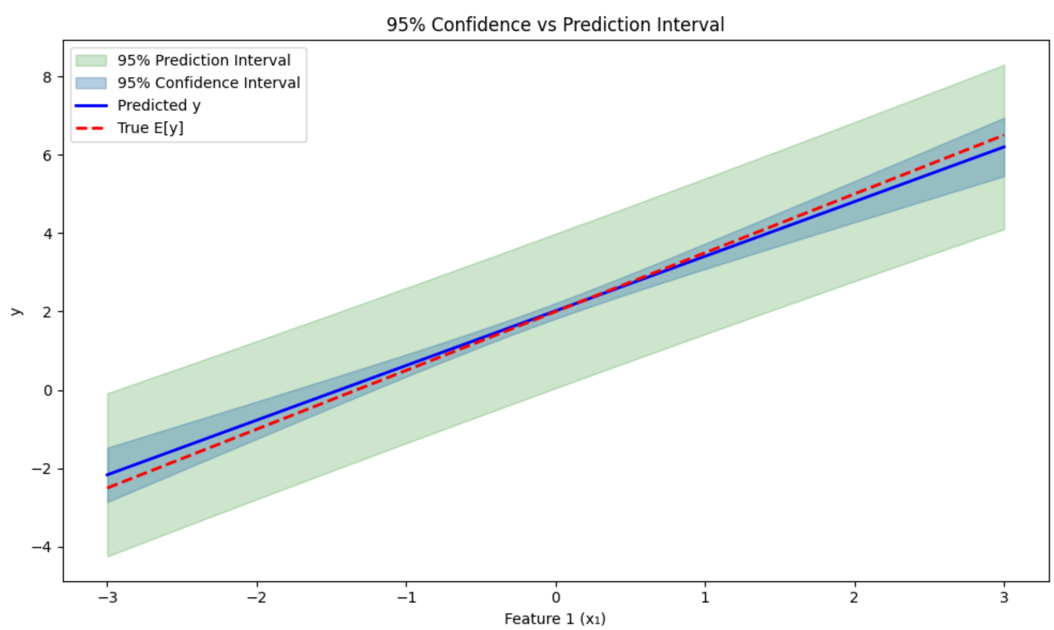

Figure 3 — 95% CI (blue, narrow) vs 95% PI (green, wide) across the feature range. Both widen at the extremes where training data is sparse.

3.5 Real Data Application — Longley Dataset

Dataset: US economic data 1947–1962 (n=16). Response y = TOTEMP (total employment). Features: GNP, UNEMP, POP.

3.5.1 OLS Results

Coefficient	OLS Estimate	OLS Std	WLS Estimate	WLS Std
Intercept	66,157.75	23,855.70	63,864.78	11,472.36
GNP	0.0476	0.0170	0.0459	0.0082
UNEMP	−0.3854	0.3315	−0.4075	0.1585
POP	−0.1539	0.2664	−0.1282	0.1280

$R^2 = 0.9812$ (OLS) = 0.9812 (WLS). WLS std errors are 52% smaller. Efficiency: OLS trace = 5.69×10^8, WLS trace = 1.32×10^8 ⇒ WLS is 4.33× more efficient.

3.5.2 Residual Analysis and Heteroscedasticity

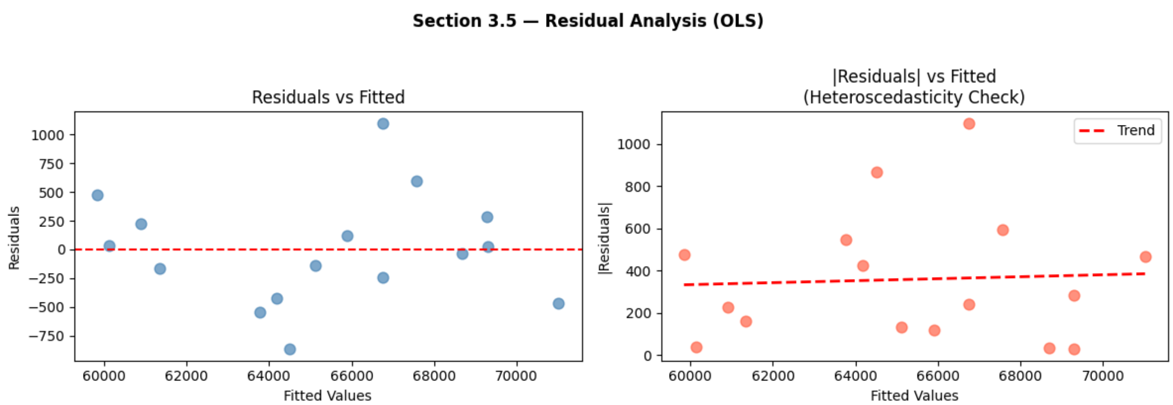

Figure 4 — Left: Residuals vs Fitted (no clear pattern ⇒ no heteroscedasticity). Right: |Residuals| vs Fitted with nearly flat trend.

Breusch-Pagan Test: p-value = 0.7550 > 0.05 ⇒ No significant heteroscedasticity ⇒ OLS is adequate. WLS still applied for comparison, yielding smaller standard errors confirming WLS ≥ OLS always.

Bonus — Fisher Information, CRLB, and Sufficient Statistics

B.1 Fisher Information Matrix and CRLB

The Fisher Information Matrix for the Gaussian linear model:

$$I(\beta) = (1/\sigma^2)\ X^TX$$

The Cramér-Rao Lower Bound states that for any unbiased estimator:

$$Var(\hat{\beta}) \geq I(\beta)^{-1} = \sigma^2(X^TX)^{-1}$$

Since $Cov(\hat{\beta}\_OLS) = \sigma^2(X^TX)^{-1}$ = CRLB exactly (difference = zero matrix), OLS achieves the lower bound and is therefore the MVUE (Minimum Variance Unbiased Estimator).

Coefficient	CRLB	OLS Covariance	Achieves Bound?
β_0	0.010199	0.010199	✓
β_1	0.013790	0.013790	✓
β_2	0.010135	0.010135	✓

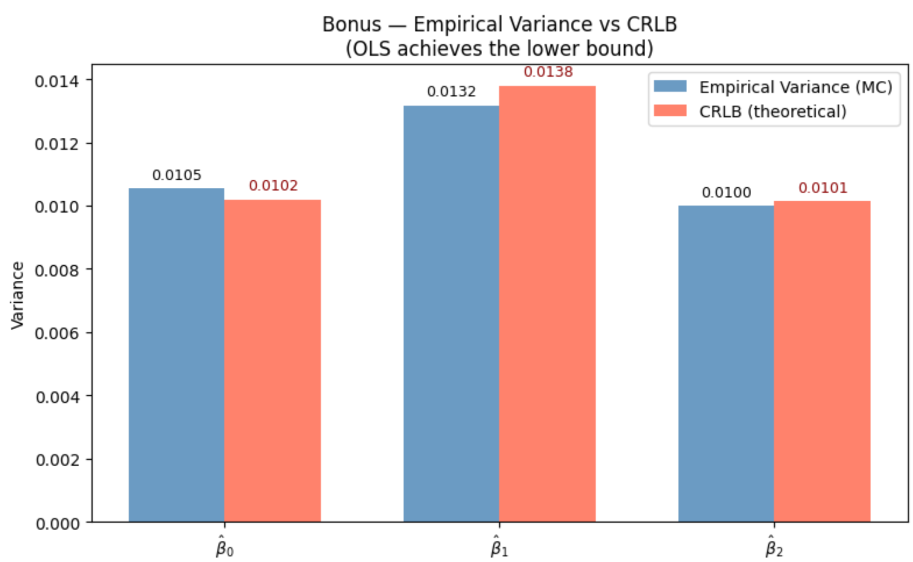

Figure 5 — Empirical variance from 5000 MC runs (blue) vs CRLB (red). Near-perfect match confirms OLS achieves the Cramér-Rao Lower Bound.

B.2 Sufficient Statistics

By the Factorization Theorem, expanding $||y - X\beta||^2 = y^Ty - 2\beta^T(X^Ty) + \beta^T(X^TX)\beta$, the likelihood depends on the data only through:

Statistic	Formula	Sufficient for	Verification
$T_1(y)$	X^Ty	β	beta from T_1 = beta from y ✓
$T_2(y)$	y^Ty	σ^2 (with T_1)	σ^2 from (T_1,T_2) = σ^2 from y ✓

$T_1 = X^Ty$ compresses 100 observations into 3 numbers with zero information loss about β. This is the theoretical basis for why OLS is the optimal estimator.